Fixed-noise weak GMM across independent records

S14. Common-trajectory GMM reference and proof

The replicate asymptotic is distinct from single-path infill and from Corollary 1. Let $Y_r = x_{\theta_0,u} + \eta_r$, $r = 1, \dots, n$, be independent copies on the same fixed interval. The deterministic trajectory and initial state are shared, the covariance is stationary Gaussian, and its amplitude is any fixed positive finite value. Assume a supplied polynomial vector field linear in $\theta \in \mathbb{R}^p$, a fixed collection of boundary-vanishing tests, and exact integrals. Write the Hermite-corrected weak moment as

$$m_r(\theta, \nu) = b_r - G_r(\nu)\theta \in \mathbb{R}^M, \qquad D = EG_r(\nu_0), \qquad Em_r(\theta_0, \nu_0) = 0,$$

where ν contains the marginal variances used in the correction and any covariance-shape parameters. The expectation identity follows by the Gaussian heat/Hermite identity at every time, Fubini, and the deterministic weak ODE identity. It is exact at fixed noise; it does not assert that the ratio or least-squares estimator from one record is unbiased.

Lemma 2 (fixed-noise independent-record GMM). Suppose D has column rank p, the covariance $\Omega = \mathrm{Cov}(m_r(\theta_0, \nu_0))$ is positive definite, and moments have finite second moments. With known ν_0 and random positive-definite $W_n \to_p W$, define

$$\hat{\theta}_n = (\bar{G}_n^\top W_n \bar{G}_n)^{-1} \bar{G}_n^\top W_n \bar{b}_n, \qquad P_W = (D^\top W D)^{-1} D^\top W.$$

Then $\hat{\theta}_n \to_p \theta_0$ and

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, V_W), \qquad V_W = P_W \Omega P_W^\top.$$

Within this fixed moment family and the corresponding regular linear GMM influence class, $W = \Omega^{-1}$ minimizes covariance, giving $V_* = (D^\top \Omega^{-1} D)^{-1}$. For polynomial moments, Ω is the full covariance from Proposition 3, including every nonzero Hermite order. Empirical residual covariance at a consistent preliminary estimate, or a consistent full model-based covariance, supplies feasible two-step weights. First-order covariance alone generally gives a different, nonoptimal V_W at fixed noise. The theorem does not assert full-path likelihood efficiency at finite amplitude.

For estimated ν, additionally assume local differentiability and an asymptotically linear nuisance estimator

$$\sqrt{n}(\hat{\nu} - \nu_0) = n^{-1/2} \sum_{r=1}^{n} \xi_r + o_p(1), \quad E\xi_r = 0, \quad E\|\xi_r\|^2 < \infty, \quad H = E\partial_\nu m_r(\theta_0, \nu_0).$$

Assume the corresponding sample derivative converges locally, with a uniform second-order Taylor remainder $o_p(n^{-1/2})$. Then the same closed-form estimator, evaluated at $\hat{\nu}$, is consistent and asymptotically normal with

$$\psi_r = m_r(\theta_0, \nu_0) + H\xi_r, \quad \widetilde{\Omega} = \mathrm{Cov}(\psi_r), \quad \widetilde{V}_W = P_W \widetilde{\Omega} P_W^\top.$$

If $\widetilde{\Omega}$ is positive definite, estimated inverse $\widetilde{\Omega}$ is optimal **for this specified nuisance estimator and moment influence family**. Its optimum is not generally the known-covariance optimum or the globally optimal joint GMM estimator. Estimating only a shape parameter that changes weights but not mean correction has zero H for that parameter; estimating a variance used in Hermite correction need not. A consistent empirical influence covariance, including cross-covariances between m_r and ξ_r, yields marginal Wald intervals and joint ellipsoids with asymptotic nominal coverage. These are standard GMM and two-step-estimation arguments specialized to the weak Hermite moments [1, 2].

Proof. Polynomial Gaussian functionals on a bounded interval with continuous deterministic mean have moments of all finite orders; bounded test functions allow the required exchanges of integration and expectation. For known covariance, the law of large numbers gives $\bar{G}_n \to D$, $\bar{b}_n \to D\theta_0$, and convergence of the inverse normal matrix. The exact identity

$$\hat{\theta}_n - \theta_0 = P_n \bar{m}_n(\theta_0, \nu_0)$$

and the iid multivariate CLT give the stated limit, without a small-noise expansion. If $PD = I_p$, set $P_* = (D^\top \Omega^{-1} D)^{-1} D^\top \Omega^{-1}$. Then

$$P\Omega P^\top - P_* \Omega P_*^\top = (P - P_*)\Omega(P - P_*)^\top \succeq 0,$$

because the cross terms vanish. A consistent preliminary least-squares estimate makes the empirical covariance converge to Ω by the law of large numbers and polynomial continuity. In the nuisance case, the stipulated Taylor expansion gives

$$\sqrt{n}\bar{m}_n(\theta_0, \hat{\nu}) = n^{-1/2} \sum_r [m_r(\theta_0, \nu_0) + H\xi_r] + o_p(1).$$

The random design and random weights multiply a $O_p(n^{-1/2})$ centered mean, so their consistency suffices for the leading map. Apply the joint CLT to (m_r, ξ_r) and the same covariance decomposition using $\widetilde{\Omega}$. Consistency of the sandwich and coverage follow by Slutsky. This proves the assertions.

Numerical quadrature condition. On a fixed discrete grid, Hermite correction is still unbiased for the corresponding *quadrature moment*. Its deterministic expectation is the weak quadrature defect, which is generally nonzero at θ_0. Consistency for the continuous ODE therefore requires exact integration or vanishing defect. In a triangular array, require $\sqrt{n}\|E m_{r,h_n}(\theta_0, \nu_0)\| \to 0$, convergence of designs and covariance, and uniform integrability sufficient for the triangular CLT and nuisance remainder. Mean-square convergence of quadrature moments to their continuous versions gives the stochastic part for this fixed test family. This condition concerns grid refinement with independent records; it does not prove a joint growing-window/small-noise rate. The numerical experiment uses a fixed fine grid and discloses its clean defect rather than claiming literal fixed-grid consistency for the ODE parameter.

S14.1. Consistent OU shape from independent copies

For even $n = 2B$, define independent pairs $d_r = (Y_{2r-1} - Y_{2r})/\sqrt{2}$. Their common deterministic mean cancels and d_r has the original stationary OU covariance. At a fixed nondegenerate grid $t_0, \ldots, t_L$ with mesh $h > 0$, let

$$s_{rj} = \frac{1}{L+1}\sum_{i=0}^{L} d_{rj}(t_i)^2, \quad a_r = \sum_{j,i<L} d_{rj}(t_i)d_{rj}(t_{i+1}), \quad c_r = \sum_{j,i<L} d_{rj}(t_i)^2.$$

Then $\hat{v}_j = B^{-1}\sum_r s_{rj} \to v_j$, $\hat{\rho} = \bar{a}/\bar{c} \to e^{-h/\tau}$, and $\hat{\tau} = -h/\log\hat{\rho} \to \tau$. Their joint $\sqrt{B}$ normality follows from the independent-pair CLT and the delta method; dependence of time points is retained inside each pair. Positivity of variances and $0 < \rho < 1$ make the transformation regular. The influence for v_j is $s_{rj} - v_j$, and for ρ it is $(a_r - \rho c_r)/Ec_r$; the τ influence is $h\xi_{\rho,r}/[\rho(\log\rho)^2]$. If finite-sample clipping is needed, any fixed bounds containing the true parameters in their interior are asymptotically inactive and must be reported.

Use pair-averaged weak moments for inference, so the theorem applies with sample size B, independent pair influences, and variance divided by $B = n/2$. In Van der Pol the only correction derivative is $H_{v_1,2k} = \theta_4 E\int \phi_k Y_2$; the other rows and variance/shape derivatives are zero. Thus the empirical covariance of pair residuals plus this derivative times $(s_{r1} - \hat{v}_1)$ supplies the nuisance-adjusted weights and sandwich. Averaging pair differences for shape estimation requires a common deterministic mean; heterogeneous initial conditions would require a different construction. Replication permits separate OU variance and range consistency that is unavailable under fixed-horizon infill of one observed path.

S14.2. Why stacking can retain bias

Stacked corrected least squares converges, when its normal matrix is nonsingular, to

$$\theta_{\text{stack}} = \theta_0 + \{E(G_r^\top G_r)\}^{-1}E[G_r^\top m_r(\theta_0, \nu_0)].$$

Exact $Em_r = 0$ does not imply $E[G_r^\top m_r] = 0$, since the design and residual share measurement error. Stacking is therefore generally biased and inconsistent in this errors-in-variables setting; special orthogonality exceptions are possible. Averaged-moment GMM instead uses $\bar{G}_n^\top W_n \bar{m}_n$, whose probability limit is zero at the true parameter. The stacking behavior in Table S11 is an instance of this distinction, not a claim that every stacked regression must be inconsistent.

With known covariance, the same argument extends to iid random initial states sharing the coefficient vector, provided the conditional weak identity, full-rank mean design and finite moments hold. This covers the mechanism in the heterogeneous-initial-state aggregation example. The pair-difference nuisance estimator above still requires a common deterministic mean and is not applied to that heterogeneous example.

S14.3. Misspecified temporal weighting

For known marginal correction and any nonsingular assumed temporal shape, the fixed-family variance is the sandwich $P_W\Omega_{\text{true}}P_W^\top/n$. In the small-noise single-record setting replace Ω_{true} with the order-one covariance and the factor $1/n$ with ϵ^2. Correct marginal correction preserves averaged-moment consistency even if τ is wrong; a wrong variance used for Hermite correction can destroy the moment identity. The finite-noise fitted-shape rows in S12 change both correction and weights and cannot be read as a pure temporal-weighting experiment. The $\tau/2$ and 2τ rows retain the true marginal variances.

References

[1] Hansen LP. Large sample properties of generalized method of moments estimators. *Econometrica* 50 (1982) 1029–1054. https://doi.org/10.2307/1912775.

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